

Stochastic Process

Definition: A collection of random variables $\{X_t\}_{t \in T}$

Geometric series: $\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}, |r| < 1$

Generalized Binomial Theorem: $(x+y)^{-n} = \sum_{k=0}^{\infty} C_{n+k-1}^k (-x)^{-k} y^{-n-k}$

Transformation of random variables If $f_Y(y) = f_X(h(y))$ and $f_Y(y) = f_X(h(y)) |\det(J_h(y))|$ 其中我们有
$$\begin{pmatrix} \frac{\partial h_1}{\partial y_1} & \cdots & \frac{\partial h_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_n}{\partial y_1} & \cdots & \frac{\partial h_n}{\partial y_n} \end{pmatrix}.$$

Bernoulli Process

Bernoulli distribution: $X \sim \text{Bern}(p)$, pmf: $P(X=1) = p, P(X=0) = 1-p$, mgf: $M_X(t) = pe^t + 1-p$

Bernoulli Process A sequence $\{X_1, X_2, \dots\}$ of independent Bernoulli variables with $X_i \sim \text{Bern}(p)$

Number of success by time n : $S_n = X_1 + \dots + X_n$

Binomial distribution: $S_n \sim B(n, p)$ pmf: $P(S_n = k) = C_n^k p^k (1-p)^{n-k}$, for $k = 0, \dots, n$, mgf: $M_{S_n}(t) = (pe^t + 1-p)^n$

k -th time of success: $T_k = \min\{n : S_n = k\}$

Geometric Distribution: represents the **first success time** in a Bernoulli process, that is $T_1 \sim \text{Geo}(p)$. pmf:

$$P(T_1 = n) = (1-p)^{n-1} p, E[T_1] = \frac{1}{p}, \text{Var}[T_1] = \frac{1-p}{p^2} \text{ and } M_{T_1}(t) = \frac{pe^t}{1-(1-p)e^t}$$

Negative binomial distribution: represents the **r -th success time**, that is $T_r \sim \text{NB}(r, p)$. pmf: $P(T_r = n) = C_{n-1}^{r-1} (1-p)^{n-r} p^r$.

$$E[T_r] = \frac{r}{p}, \text{Var}[T_r] = r \frac{1-p}{p^2}, M_{T_r}(t) = \left(\frac{pe^t}{1-(1-p)e^t} \right)^r$$

i.i.d. waiting times between successes: Denoting $W_i = T_i - T_{i-1}$, Then $\{W_1, W_2, \dots\}$

$W_2, \dots, W_k = \{W_i : i=1, \dots, k\}$ (memoryless property)

Law of large number: Frequency and probability. $\lim_{n \rightarrow \infty} \frac{S_n}{n} = p$

For discrete case: $\mathbb{E} \left[\lim_{n \rightarrow \infty} \frac{S_n}{n} \right] = \lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_n}{n} \right] = p$ and

$$\text{Var} \left[\lim_{n \rightarrow \infty} \frac{S_n}{n} \right] = \lim_{n \rightarrow \infty} \text{Var} \left[\frac{S_n}{n} \right] = 0$$

Poisson Process

From Bernoulli (Counting Process) to Poisson:

考虑到时间 k 成功的次数 S_k , 对于 Bernoulli 只会在整数的时间点更新 $\Rightarrow$ 更加频繁的做实验, k/m 时间里做 k 个实验, $S_{k/m}^{(m)} \sim B(k, p/m)$

$\Rightarrow m \rightarrow \infty$, limiting process S_t^{lim} , 取 $t=1$ 符合 Poisson 分布。

$$\begin{aligned} \mathbb{P}(S_1^{lim} = k) &= \lim_{m \rightarrow \infty} \mathbb{P}(S_1^{(m)} = k) = \lim_{m \rightarrow \infty} C_m^k \left(\frac{p}{m} \right)^k \left(1 - \frac{p}{m} \right)^{m-k} \\ &= \lim_{m \rightarrow \infty} \frac{m!}{(m-k)! k!} \cdot \frac{p^k}{m^k} \left(1 - \frac{p}{m} \right)^m \cdot \left(1 - \frac{p}{m} \right)^{-k} \\ &= \frac{p^k}{k!} \lim_{m \rightarrow \infty} \left(1 - \frac{p}{m} \right)^m \cdot \frac{m(m-1) \cdots (m-k+1)}{(m-p)^k} \\ &= \frac{p^k}{k!} e^{-p}. \end{aligned}$$

Poisson Distribution: $X \sim \text{Pois}(\lambda)$, $P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}$, for $k \geq 0$

$$E[X] = \lambda, \text{Var}(X) = \lambda \text{ and } M_X(t) = e^{\lambda(e^t-1)}$$

Properties 1 (Independent Increment): $S_t^{(m)} - S_s^{(m)} \perp \{S_r^{(m)}\}_{r \leq s}$

Properties 2 (Stationary Increment): $S_t^{(m)} - S_s^{(m)} \stackrel{d}{=} S_{t-s}^{(m)}$

Definition:

A counting process $\{N_t\}_{t \geq 0}$ is called a Poisson process with rate $\lambda > 0$ if it satisfies the following properties: for all $s \leq t, 1. N_0 = 0$

2. independent increment: $N_t - N_s \perp N_r$ for $r < s$. 3. stationary increments: $N_t - N_s \sim \text{Pois}(\lambda(t-s))$.

First arrival time: $T_1 \sim \text{Exp}(\lambda)$

Exponential Distribution:

cdf: 通过“直到时间 t 都没有事件发生”这一逻辑推导: $F_{T_1}(t) = \mathbb{P}(T_1 \leq t) = \mathbb{P}(N_t > 0) = 1 - e^{-\lambda t}$, 对于 $t \geq 0$.

pdf: 对 cdf 求导得到 $f_{T_1}(t) = \lambda e^{-\lambda t}$, 对于 $t \geq 0$.

Properties: $E[T_1] = \frac{1}{\lambda}$, $\text{Var}(T_1) = \frac{1}{\lambda^2}$ 且 $M_{T_1}(t) = \frac{\lambda}{\lambda - t}$.

Memoryless property: $\mathbb{P}(T_1 > t+s | T_1 > s) = \mathbb{P}(T_1 > t)$. 这意味着已经过去的时间不会影响未来事件发生的概率, 这一性质与离散情形下的 $\text{Geo}(p)$ 一致。

r -th arrival time: $T_r \sim \text{Gamma}(r, \lambda)$

我们在研究泊松过程中第 r 个事件发生的时间 T_r 。这可以看作是 r 个独立的、参数为 λ 的等待时间 (即间隔 W_i) 的总和。

Gamma Distribution (Erlang Distribution):

cdf: $F_{T_r}(t) = \mathbb{P}(N_t \geq r) = 1 - \sum_{i=0}^{r-1} \frac{(\lambda t)^i}{i!} e^{-\lambda t}$, 对于 $t \geq 0$.

pdf: 对 cdf 求导得到 $f_{T_r}(t) = \frac{\lambda^r t^{r-1} e^{-\lambda t}}{(r-1)!}$, 对于 $t \geq 0$.

Properties: $E[T_r] = \frac{r}{\lambda}$, $Var(T_r) = \frac{r}{\lambda^2}$ 且 $M_{T_r}(t) = \left(\frac{\lambda}{\lambda-t}\right)^r$ 。这与伯努利过程中的负二项分布 $NB(r, p)$ 形成直接对应。

Allow r to take non-integer values: 定义 $\Gamma(r) := \int_0^\infty t^{r-1} e^{-t} dt$

其中 $\Gamma(r) = (r-1)!$, $\Gamma(r+1) = r\Gamma(r)$, $\Gamma(1) = 1$ $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

$f_X(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}$ 并且 $E[T_r] = \frac{r}{\lambda}$ $Var(T_r) = \frac{r}{\lambda^2}$ 和 $M_{T_r}(t) = \left(\frac{\lambda}{\lambda-t}\right)^r$

i.i.d. waiting times between arrivals:

1. Joint distribution

$\mathbb{P}(T_1 < t_1, T_2 < t_2) = \mathbb{P}(N(t_1) > 0, N(t_2) > 1)$ if $t_1 \geq t_2$ 那么

$\mathbb{P}(N(t_1) > 0, N(t_2) > 1) = \mathbb{P}(N(t_2) > 1) = 1 - \mathbb{P}(N(t_2) = 0) - \mathbb{P}(N(t_2) = 1) = 1 - e^{-\lambda t_2} - \lambda t_2 e^{-\lambda t_2}$

if $t_1 < t_2$ 那么

$= \mathbb{P}(N(t_1) = 1, N(t_2) > 1) + \mathbb{P}(N(t_1) > 1) = \mathbb{P}(N(t_1) = 1, N(t_2) - N(t_1) > 0) + \mathbb{P}(N(t_1) > 1)$

$= \mathbb{P}(N(t_1) = 1) \cdot \mathbb{P}(N(t_2) - N(t_1) > 0) + 1 - \mathbb{P}(N(t_1) = 0) - \mathbb{P}(N(t_1) = 1) = \lambda t_1 e^{-\lambda t_1} (1 - e^{-\lambda(t_2-t_1)}) + 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_1}$

$= 1 - e^{-\lambda t_2} - \lambda t_2 e^{-\lambda t_2}$

pdf: $f_{T_1, T_2}(t_1, t_2) = \lambda^2 e^{-\lambda t_2}$, $0 < t_1 < t_2$

2. Waiting time i.i.d.:

$f_{T_1, T_2 - T_1}(t_1, t_2) = f_{T_1, T_2}(t_1, t_1 + t_2) = \lambda^2 e^{-\lambda(t_1+t_2)}$, 对于 $t_1, t_2 > 0$.

$f_{T_1}(t_1) = \lambda e^{-\lambda t_1}$ 和 $f_{T_2 - T_1}(t_2) = \lambda e^{-\lambda t_2}$, 对于 $t_1, t_2 > 0$

Principle of superposition: If $\{N^{(1)}\}_t, \dots, \{N^{(k)}\}_t$ are independent Poisson processes with rates $\mu_1, \dots, \mu_k$, then $\{N^{(1)} + \dots + N^{(k)}\}_t$ is a Poisson process with rate $\mu_1 + \dots + \mu_k$.

Principle of thinning: If N_t is a Poisson process with rate μ , and if each arrival is independently assigned one of k labels with probabilities $p_1, \dots, p_k$, then the counting processes $\{N_t^{(1)}\}, \dots, \{N_t^{(k)}\}$ that count the arrivals of each type are independent Poisson processes with rates $p_1\mu, \dots, p_k\mu$.

Simple Random Walk and Brownian Motion

1. Simple Random Walk (SRW)

一维随机游走描述了一个质点在直线上依概率随机移动的过程。

Definition: 随机游走 $\{S_n\}_{n \geq 0}$ 定义为 $S_n = S_0 + X_1 + \dots + X_n$, 其中 X_i 是独立同分布 (i.i.d.) 的随机变量。

Symmetric SRW: 当 S_0 为整数且步长 $X_i \in \{1, -1\}$, 且 $P(X_i = 1) = P(X_i = -1) = \frac{1}{2}$ 时, 称为对称简单随机游走。

Properties: 若 $S_0 = 0$, 则对于对称 SRW 有:

$\mathbb{E}[S_n] = 0$ $Var(S_n) = n$

2. Continuous Time Limit (From SRW to BM)

为了得到连续时间下的过程, 我们对步长和时间间隔进行缩放。Mean和Variance要和原来的一致

Scaling: 考虑在单位时间内进行 N 次实验, 为了保持方差不发散, 步长需缩放为 $\frac{1}{\sqrt{N}}$ 。定义缩放后的过程为:

$$S_t^{(N)} = \frac{1}{\sqrt{N}} \sum_{i=1}^{\lfloor Nt \rfloor} X_i^{(N)}$$

Limiting Distribution: 当 $N \rightarrow \infty$ 时, 根据矩母函数 (mgf) 推导:

$$M_{S_1^{(N)}}(t) = \left(\frac{e^{t/\sqrt{N}} + e^{-t/\sqrt{N}}}{2} \right)^N \rightarrow e^{t^2/2}$$

这表明 $S_1^{(N)}$ 在极限下服从标准正态分布 $N(0, 1)$ 。

3. Brownian Motion (BM)

Definition: 一个连续过程 $\{B_t\}_{t \geq 0}$ 称为标准布朗运动, 若其满足:

$B_0 = 0$ 。

Independent Increments: 对于任意 $s \leq t$, 增量 $B_t - B_s$ 独立于历史过程 $\{B_r\}_{r \leq s}$ 。

Stationary Increments: 增量 $B_t - B_s \sim N(0, t-s)$ 。

Joint Distribution: 布朗运动在不同时间点 $(B_{t_1}, B_{t_2}, \dots, B_{t_n})$ 的联合分布是联合正态分布 (Jointly Normal), 其均值为 0, 协方差矩阵 Σ 满足 $\Sigma_{ij} = Cov(B_{t_i}, B_{t_j}) = \min(t_i, t_j)$ 。

PDF 中的协方差矩阵表示为 $Cov(B_{t_i}, B_{t_j}) = t_i$ (若 $t_i < t_j$)。

4. Normal Distribution & Central Limit Theorem (CLT)

布朗运动的理论基础是正态分布的普适性。

Normal Distribution: $X \sim N(\mu, \sigma^2)$ 的 pdf 为 $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ 。

其 mgf 为 $M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$ 。

Joint Normal Distribution: $X = (X_1, X_2, \dots, X_n)$ 是联合正态分布如果 $\sum_{i=1}^n a_i X_i$ 对于所有 $a_1, \dots, a_n \in \mathbb{R}$ 都是正态分布。记作 $\mathbf{X} \sim \mathbf{N}(\mu, \Sigma)$, 其中 $\mathbb{E}[\mathbf{X}] = \mu$, 协方差矩阵为 Σ 。

$$f_{B_{t_1}, B_{t_2}-B_{t_1}, \dots, B_{t_n}-B_{t_{n-1}}}(x_1, \Delta x_2, \dots, \Delta x_n) = \frac{1}{\sqrt{2\pi t_1}} e^{-\frac{x_1^2}{2t_1}} \frac{1}{\sqrt{2\pi(t_2-t_1)}} e^{-\frac{\Delta x_2^2}{2(t_2-t_1)}} \dots \frac{1}{\sqrt{2\pi(t_n-t_{n-1})}} e^{-\frac{\Delta x_n^2}{2(t_n-t_{n-1})}}$$

Characterization of Brownian Motion: 对于 $t_1 \leq t_2 \leq \dots \leq t_n$, 有:

$$f_{B_{t_1}, B_{t_2}, \dots, B_{t_n}}(x_1, x_2, \dots, x_n) = \frac{1}{\sqrt{(2\pi)^n t_1(t_2-t_1) \dots (t_n-t_{n-1})}} \exp\left(-\frac{x_1^2}{2t_1} - \frac{(x_2-x_1)^2}{2(t_2-t_1)} - \dots - \frac{(x_n-x_{n-1})^2}{2(t_n-t_{n-1})}\right)$$

Central Limit Theorem: 设 $\{X_k\}$ 为 i.i.d. 随机变量, 且具有有限均值 μ 和方差 σ^2 。当 $n \rightarrow \infty$ 时, 其样本均值 $\bar{X}$ 的标准化形式收敛于标准正态分布:

$$W = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

这意味着无论原始分布如何, 大量独立随机变量之和 (经过适当缩放) 总趋向于正态分布。